

一、前置知识:

1. 排列组合 ~

2. 斯特林公式: $\ln(n!) \approx n \ln n - n$ ($m! = m^m e^{-m} \sqrt{2\pi m} (1 + \frac{1}{12m} + \frac{1}{288m^2} + \dots)$)
stirling 快速收敛 $m \rightarrow \infty$

3. 热一律: 1) $du = dQ + dW = Tds - pdv$
 2) $pV = nRT = NkT$ $k = 1.38 \times 10^{-23} \text{ J/K}$

二、麦-玻分布

1. 建模: 可辨粒子群

态	i
能级	ϵ_i
简并度	w_i
粒子数	a_i (确定的)

空间中:	体积元 Δw_i
能级	ϵ_i
粒子数	a_i
体积元	$\Delta w_i = \prod_{j=1}^r \Delta q_j \prod_{j=1}^s \Delta p_j$
简并度	$\frac{\Delta w_i}{h_0^r}$

$$\textcircled{1} \quad \sum_i a_i = N \quad \sum_i a_i \epsilon_i = E$$

$h_0 = \bar{q} \cdot \bar{p}$ (最小单位)

量子 $h_0 = h$, 经典 $h_0 \rightarrow 0$
近似为连续简并

状态数

② $\Omega_{M.B} \rightarrow$ 有号球自由入袋

① ② ③ ... ② 盒子无限装, 小球随意进

当前状态数: $\Omega_{M.B} = \frac{N!}{\prod_i a_i!} \prod_i w_i^{a_i}$

$\xleftrightarrow[\varepsilon_1]{w_1} \quad \xleftrightarrow[\varepsilon_2]{w_2} \quad \xleftrightarrow[\varepsilon_3]{w_3} \rightarrow P = \frac{1}{\Omega}$

2. 最概然分布:

对 $\Omega_{M.B}$ 分析: $\ln \Omega = \ln N! - \sum \ln a_i! + \sum a_i \ln w_i$

$\approx N \ln N - N - \sum a_i (\ln a_i - a_i) + \sum a_i \ln w_i$

$= N \ln N - \sum a_i \ln a_i + (\sum a_i - N) + \sum a_i \ln w_i$

变分: $a_i \pm \delta a_i \rightarrow$ 1) $\sum \delta a_i = \delta N = 0$ 2) $\sum \varepsilon_i \delta a_i = \delta E = 0$
 a_i 很大, 可不看作整数

$(x \ln x)' = \ln x + 1$ $\delta \ln \Omega = -(\sum \delta a_i (\ln a_i + 1)) + \sum \delta a_i \ln w_i$
 $= -\sum \delta a_i \ln \frac{a_i}{w_i}$

拉乘子法: $\delta \ln \Omega + \alpha \delta N + \beta \delta E = 0$

$\downarrow$
 $(\sum \ln \frac{a_i}{w_i} + \alpha + \varepsilon_i \beta) \delta a_i = 0$

$\ln \frac{a_i}{w_i} + \alpha + \varepsilon_i \beta = 0 \rightarrow \underline{a_i = w_i e^{\alpha - \beta \varepsilon_i}}$

之后, 我们还要确定该极值为极大, 且最概然:

极大: $\delta^2 \ln \Omega = -\delta(\sum \ln \frac{a_i}{w_i} \cdot \delta a_i) = -\sum \frac{(\delta a_i)^2}{a_i} < 0$, 得证

$N = \sum e^{\alpha - \beta \varepsilon_i}$
 $E = \sum \varepsilon_i e^{\alpha - \beta \varepsilon_i}$

$\rightarrow Z = \sum_i a_i \cdot e^{-\alpha}$
 $= \sum_i w_i e^{-\beta \varepsilon_i}$

★ 粒子配分函数

最概：(看偏移值)

$$\ln(\Omega + \Delta\Omega) = \ln\Omega + \delta \ln\Omega + \frac{1}{2} \delta^2 \ln\Omega - \dots$$

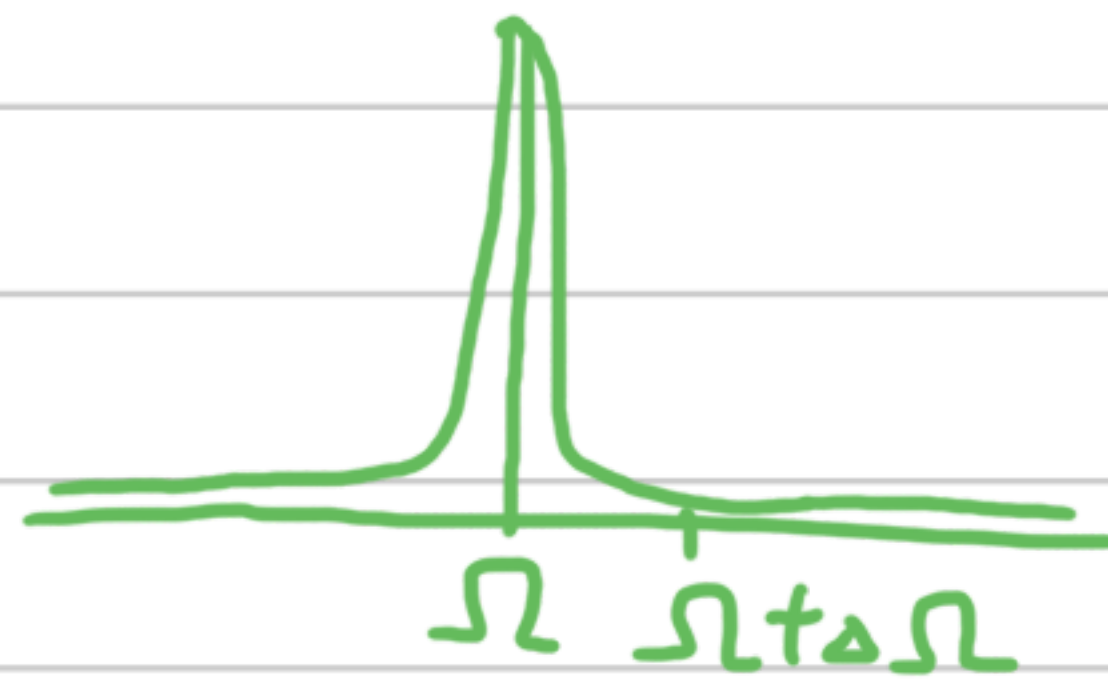
↓

$$\ln(\Omega + \Delta\Omega) - \ln\Omega = -\frac{1}{2} \sum \frac{\partial a_i^2}{a_i^2}$$

(对于近独立粒子, 粒子的偏移等概)

对 $N \approx 6 \times 10^{23}$, 若 $\frac{\partial a_i}{a_i} \approx 10^{-5}$, $\frac{\Omega + \Delta\Omega}{\Omega} \approx e^{-\frac{1}{2} \cdot (10)^3}$

a_i 对 Ω 影响极小, 因此在有微小偏差的情况下依然可用这个分布



3. 配分函数推热力学量: $Z = \sum w_i e^{-\beta \epsilon_i}$

1) 粒子数 $N = e^{-\alpha} \cdot Z$

2) 内能 $U = E = e^{-\alpha} \sum \epsilon_i w_i e^{-\beta \epsilon_i} = e^{-\alpha} \cdot \frac{\partial}{\partial \beta} (-Z) = -N \cdot \frac{\partial}{\partial \beta} \ln Z$

3) 广义力 $Y_a = \sum_{\text{粒子 } i} \frac{\partial \epsilon_i}{\partial q_a} = \sum_{\text{粒子 } i} \frac{\partial \epsilon_i}{\partial q_a} a_i = \sum e^{-\alpha} \frac{\partial \epsilon_i}{\partial q_a} \cdot w_i e^{-\beta \epsilon_i} = e^{-\alpha} \cdot \left(-\frac{1}{\beta} \frac{\partial}{\partial q_a}\right) \sum w_i e^{-\beta \epsilon_i}$

$$= \frac{N}{Z} \left(-\frac{1}{\beta} \frac{\partial}{\partial q_a}\right) Z$$

$$= -\frac{N}{\beta} \cdot \frac{\partial \ln Z}{\partial q_a}$$

3.5) 压强: $T ds - p dv = du \Rightarrow p = -Z \frac{\partial \epsilon_i}{\partial v} a_i = \frac{N}{\beta} \frac{\partial \ln Z}{\partial v}$

or $C dT$
 $-\beta \frac{\partial}{\partial \beta} (\ln Z) \cdot \frac{dT}{T}$

4) 温度: $dQ = T dS = dU - dW = -N d\left(\frac{\partial \ln Z}{\partial \beta}\right) + \frac{N}{\beta} \frac{\partial \ln Z}{\partial q} dq$
 $= \frac{1}{\beta} d\left(N \left(\ln Z - \beta \frac{\partial \ln Z}{\partial \beta}\right)\right)$

$$d \ln Z = \frac{\partial}{\partial \beta} (\ln Z) d\beta + \sum \frac{\partial}{\partial q_i} (\ln Z) dq_i$$

or $-\frac{\partial}{\partial v} \ln Z dv$

↓
 $\frac{1}{\beta} = kT$

$$d \frac{1}{kT} = \frac{-1}{kT^2} dT$$

5) 熵: 由上: $dS = Nk d(\ln z - \beta \frac{\partial}{\partial \beta} \ln z) \rightarrow S = Nk (\ln z - \beta \frac{\partial}{\partial \beta} \ln z) + 0$ 定态
↓

1) $\ln z = \ln N + \alpha$

$= Nk (\ln N + \alpha + \beta \frac{U}{N})$

2) $\beta \frac{\partial}{\partial \beta} \ln z = \frac{-U}{N} \cdot \beta$

$= k (N \ln N + N\alpha + \beta U)$

3) $a_i = w_i e^{\alpha - \beta \epsilon_i}$
 $\alpha + \beta \epsilon_i - \ln \frac{w_i}{a_i} = 0$

$= k (N \ln N + \sum \alpha \cdot a_i + \sum \beta \epsilon_i a_i)$

$= k (N \ln N + \sum a_i \ln a_i - \sum a_i \ln w_i)$

$= k \ln \Omega$

4. 积分形式: $w_i = \frac{\Delta w_i}{h^3}$,

$Z = \int w_i \cdot e^{-\frac{\epsilon}{kT}} = \frac{1}{h^3} \int_{V^2} e^{-\frac{\epsilon}{kT}} dp_x dp_y dp_z \cdot dx dy dz$

理想气体: $\epsilon = \frac{1}{2m} \sum p_i^2 \rightarrow Z = \frac{1}{h^3} \int dV \cdot \left(\int_{-\infty}^{\infty} e^{-\frac{\beta}{2m} p^2} dp \right)^3$
 $= \frac{1}{h^3} V \cdot \left(\frac{2\pi m}{\beta} \right)^{\frac{3}{2}} = V \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}}$

$p = \frac{N}{\beta} \frac{\partial}{\partial V} \ln z = \frac{NkT}{V} \Rightarrow k = 1.38 \times 10^{-23} \text{ J/K}$

经典极限条件: $e^{\alpha} = \frac{Z}{N} = \frac{V}{N} \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}} \gg 1$ C 代表非简并, $Z \gg N$

德布罗意波 $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m kT}}$ $n \lambda^3 \ll 1$ (自由)
 $n = \frac{N}{V}$